

Nguyen Thanh Long

Backend / Platform Engineer

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SUMMARY

Backend engineer, 2.5 years, currently the sole engineer on the cloud platform of an IoT startup — the NestJS/MongoDB APIs, a self-hosted MQTT → Kafka pipeline, and the Kubernetes and AWS infrastructure underneath them. Used to owning a system end to end: designing it, deploying it, and being responsible for it in production.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, C#, Python, SQL
Backend	Node.js, NestJS, Express.js, ASP.NET Core, REST, WebSockets
Data & messaging	MongoDB (Atlas and self-hosted replica sets), PostgreSQL, Redis, MySQL, DynamoDB, Apache Kafka (Strimzi), MQTT (EMQX)
Infrastructure	Kubernetes (k0s, kubeadm, KEDA), AWS (ECS Fargate, S3, DynamoDB, Lambda, CloudWatch), Docker, Terraform, ArgoCD, Nginx
Operations	Prometheus, Grafana, Loki, GitHub Actions, GitLab CI, Jest
Auth	Firebase Auth, Keycloak, JWT, ABAC, OAuth 2.0

EXPERIENCE

Backend / Platform Engineer — Rogo Solutions JSC Jan 2024 – Present · Hanoi

A ~20-person IoT startup, where I am the only engineer on the cloud side — the API, the message pipeline, the clusters and the deploy path. It serves 640k users and 69k devices for 26 partner projects, at roughly 2.6M API requests/day and 15.5k concurrent MQTT connections, across 3 Kubernetes clusters and AWS ECS.

- Device data plane.** Designed and operate the state pipeline — EMQX MQTT into a 3-broker Strimzi Kafka cluster (RF3), then a decoding agent and a coalescing consumer that KEDA autoscales on partition lag, writing to MongoDB. Moved partner traffic onto it from a single-node broker via an external listener.
- Multi-tenancy.** Built the partner-facing surface for 26 projects: an attribute-based authorization layer (v1 → v2), per-project SMS providers, and phone + OTP sign-in.
- Operations.** Took the platform from no monitoring at all to Prometheus, Loki and Grafana across 3 clusters plus CloudWatch, so failures are diagnosed from a dashboard rather than by guessing. Run the staging → production path across 4 repositories (GitHub Actions, GitLab CI, ArgoCD), deploying to ECS, Kubernetes and the MQTT workers.
- Performance work.** The expensive problems were found by profiling, not guessing: an unindexed \$lookup that took 11 minutes and was about to force a database tier upgrade; RS256 token verification running per request inside auth middleware, which no APM was showing; a partner SDK stampeding every hour on the hour. Each was a small change once it was found.
- Codebase health.** Brought back a test suite that had become unrunnable — it exhausted memory and froze dev machines — by moving Jest 24 → 29 to transpile-only tests with a separate type-check, then gated CI on it. Also removed 1,667 lines of dead code with no behaviour change.
- Self-hosted edition.** Designed an on-prem deployment profile (Firebase, DynamoDB and S3 replaced by Keycloak, MongoDB and MinIO) with an auth migration that preserves existing logins, and validated it with a boot PoC that surfaced 10 concrete blockers before the company committed to it.

SELECTED PROJECTS

IoT Cloud MCP — TypeScript, Python, MCP SDK
A Model Context Protocol server for IoT cloud platforms, so an AI assistant can query and control devices through one standard interface instead of a bespoke integration per platform.

Novel Studio — NestJS, PostgreSQL + pgvector, LLM
A writing tool whose real problem is memory cost: a canon store, a rolling recursive summary and pgvector retrieval keep the prompt bounded at any story length, so context stays correct as the book grows but the budget does not.

MathSketch Tutor — NestJS, mathjs, Firebase, LLM
Step-by-step feedback on a handwritten maths problem, where a computer-algebra engine decides what is correct and the model may only explain where a step broke — so feedback can be wrong about wording, not about mathematics.

EDUCATION & ADDITIONAL

B.Sc. in Software Engineering — FPT University 2022 – 2026

Final-year project: a real-time multiplayer game backend (ASP.NET Core, WebSockets, MySQL) using lag compensation to keep play fair across uneven network conditions.

Certifications: Project Management Principles (UC Irvine, 2025) · Software Development Lifecycle Specialization (University of Minnesota, 2024).

Coach competitive-programming students for national Informatics examinations, weekly since 2024; freelance backend work alongside full-time employment. English: professional working proficiency.